Implicit Hamiltonian Optimal Control with Unknown Dynamics via Jacobian-Free Backpropagation

Marin Hanachowicz
ETH Zurich, D-MATH
mhanachowicz@ethz.ch

Ignacio E. Adrogué
ETH Zurich, D-PHYS
iadrogue@ethz.ch

Hector Stekelorom
ETH Zurich, D-INFK
hstekelorom@ethz.ch

François Barré
ETH Zurich, D-MATH
fbarre@ethz.ch

Abstract

We study the implicit-Hamiltonian JFB pipeline of Gelphman et al. [2025] through reproduction and extension to unknown dynamics. Reproducing it on the Almgren–Chriss liquidation problem, we diagnose and fix a contractivity failure in the natural problem formulation, validate the corrected formulation against exact closed-form references, and confirm JFB's compute and memory advantage over full autodiff. Our main contribution extends the pipeline to the setting where the dynamics are unavailable to the agent: every dynamics Jacobian required by the JFB construction is replaced by an online recursive least squares estimate fitted from rollouts, leaving the implicit fixed-point operator and JFB gradient structurally intact. We test this on Van der Pol stabilization, across variants of increasing implicitness and against oracle-Jacobian and exact-gradient (BPTT) baselines, and on a Merton portfolio problem with a transcendental optimality condition. The resulting policy matches the oracle-Jacobian variant within run-to-run variation and trails exact BPTT by under 2%. We state the corresponding descent property as a conjecture, not a theorem, and specify what its empirical failure would look like.

1 Introduction

Optimal control problems ask for a time-varying input that minimizes a cost accumulated along a dynamical system's trajectory. When the system is high-dimensional and the cost is nonlinear, computing the optimal feedback law is hard: classical grid-based methods suffer from the curse of dimensionality, and most learning-based alternatives require the optimal control to be expressible in closed form as a function of the value function's gradient. Many practically relevant problems, including inventory liquidation with nonlinear market impact, do not satisfy this requirement.

Gelphman et al. [2025][1] propose a pipeline that handles this case by treating the optimal control as the fixed point of an iterative operator and training a neural value function through that fixed point using Jacobian-Free Backpropagation [Wu Fung et al., 2022]. We study this pipeline by reproducing it on the Almgren–Chriss liquidation problem [Almgren and Chriss, 2001], a standard finance benchmark with an exact reference solution. In doing so we encounter a failure mode that the original paper does not discuss: the natural formulation of the problem causes the fixed-point solver to diverge for reasons that can be traced precisely to the structure of the adjoint equation. We identify the cause, propose a reformulation that fixes it, and validate the corrected pipeline against the exact solution. We then extend the pipeline to the setting where the dynamics are unknown, replacing the required Jacobians with online estimates, and test the result on two benchmarks.

[1] Our code, which is a clone of the author's original repository, is publicly available here

40th Conference on Neural Information Processing Systems (NeurIPS 2026).

The paper is organized as follows. Section 2 summarizes the optimal control framework and the JFB pipeline; the longer derivations are in the appendix (Sections A.4 and A.5). Section 3 positions the paper relative to prior work. Section 4 covers the Almgren–Chriss reproduction, including the contractivity diagnosis and reduced formulation. Section 5 develops the unknown-dynamics extension.

2 Background

We use standard deterministic optimal control notation. The agent chooses a control $u(t) \in U \subseteq \mathbb{R}^m$ to steer a state $z(t) \in \mathbb{R}^n$ and minimize a running-plus-terminal cost,

$$\min_{u(\cdot)} \int_0^T L(t, z(t), u(t)) \, \mathrm{d}t + G(z(T)) \quad \text{s.t.} \quad \dot{z}(t) = f(t, z(t), u(t)), \quad z(0) = x. \tag{1}$$

The Pontryagin maximum principle [Pontryagin et al., 1962] introduces a costate $p(t) \in \mathbb{R}^n$ and the Hamiltonian $H(t, z, p, u) = -p^\top f(t, z, u) - L(t, z, u)$. At an optimal trajectory the state and costate satisfy

$$\dot{z}(t) = f(t, z, u^\star), \quad z(0) = x, \tag{2}$$

$$\dot{p}(t) = -(\nabla_z f)^\top p - \nabla_z L, \quad p(T) = \nabla G(z(T)), \tag{3}$$

and the optimal control maximizes the Hamiltonian pointwise, $u^\star(t) \in \arg\max_u H(t, z(t), p(t), u)$. The adjoint equation (3) runs backward from the terminal condition and requires f analytically. The full derivation via calculus of variations is in Section A.5.

The value function $\phi(t, z)$, defined as the optimal cost-to-go from state z at time t, satisfies the Hamilton–Jacobi–Bellman equation and encodes the same information as the costate through the identity

$$p(t) = \nabla_z \phi(t, z^\star(t)) \,, \tag{4}$$

along any optimal trajectory [Fleming and Soner, 2006]. The feedback law therefore reads $u^\star(t, z) \in \arg\max_u H(t, z, \nabla_z \phi(t, z), u)$, and approximating ϕ by a neural network ϕ_θ gives access to a feedback controller without solving (2)–(3) directly.

When the maximization $\arg\max_u H$ has a closed form this approach scales well to high-dimensional problems [Ruthotto et al., 2020, Onken et al., 2023]. When it does not, because $\nabla_u H = 0$ is transcendental in u, the feedback law cannot be evaluated. Following Gelphman et al. [2025], we turn the optimality condition into a fixed-point problem: define

$$T_\theta(u; t, z) = u + \alpha \, \nabla_u H(t, z, \nabla_z \phi_\theta(t, z), u) \,, \qquad u^\star_\theta = T_\theta(u^\star_\theta; t, z), \tag{5}$$

where $\alpha > 0$ is a step size. Any interior fixed point of T_θ satisfies $\nabla_u H = 0$ by construction, and the fixed point is computed by iterating $u^{(k+1)} = T_\theta(u^{(k)}; t, z)$. This iteration converges whenever T_θ is a contraction in u, which holds when H is strictly concave in u; see Section A.4.1.

Training ϕ_θ requires differentiating the trajectory cost through the fixed point. The exact sensitivity involves a linear solve in the control dimension m at every time step, costing $O(m^3)$. Jacobian-free backpropagation [Wu Fung et al., 2022] replaces this solve by an identity approximation, reducing the cost to $O(m^2)$ and keeping memory constant across time steps. Gelphman et al. [2025] show that the resulting approximate gradient is a descent direction for the trajectory cost under the contractivity condition above, a conditioning assumption on $\partial T_\theta / \partial \theta$, and a time-regularity condition on the integrand.

3 Related work

We position this paper relative to prior learning-based optimal control methods along two axes: whether the dynamics f are known to the agent, and whether the Hamiltonian maximization admits a closed-form solution. Figure 1 summarizes the regimes covered by prior work and the setting we study.

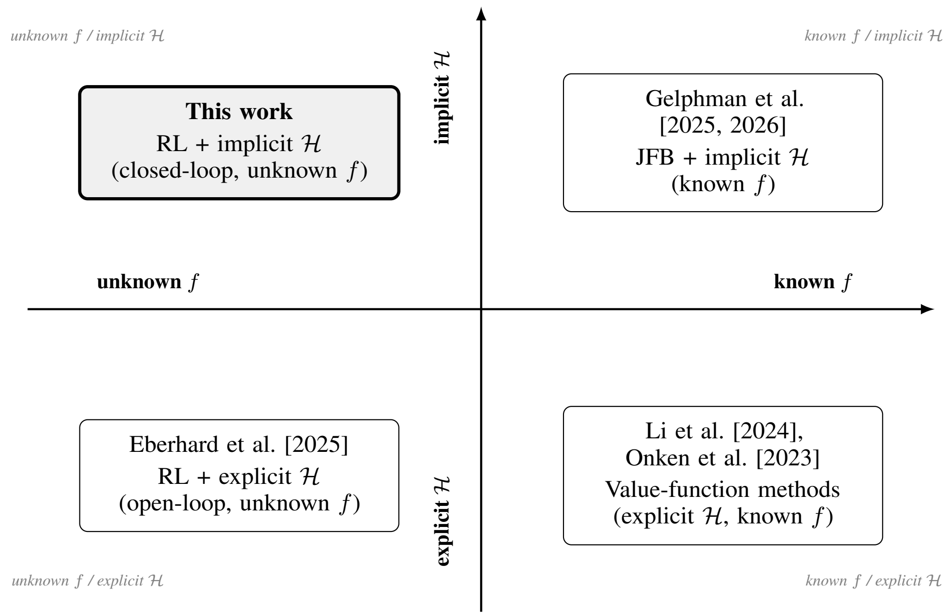


Figure 1: Positioning of related work along two axes: availability of the dynamics f to the agent (horizontal) and whether the Hamiltonian maximization $u^\star \in \arg\max_u \mathcal{H}$ admits a closed-form solution (vertical). Each prior approach covers at most one of the two challenging regimes (upper half, left half); this work is the first to address both simultaneously.

Alternative approaches to the implicit differentiation bottleneck exist in the control literature. DiffMPC [Amos et al., 2018] differentiates through a model predictive control layer via the KKT conditions of the underlying QP, giving exact gradients at the cost of a linear solve per step. IDOC [Xu et al., 2023] reduces this to linear time by exploiting the matrix equation structure directly. Learned MPC methods [Hertneck et al., 2018] replace the online solve with a neural approximation, recovering efficiency at the expense of optimal control structure. This paper follows Gelphman et al. [2025] in using JFB exclusively; a comparison of these alternative operators on implicit Hamiltonian problems is left to future work.

4 Almgren-Chriss Reproduction

4.1 Problem formulation

An agent holds q_0 shares of n risky assets and must liquidate the entire position over $[0, T]$ at minimum risk-adjusted cost. The state is $z = (q, S) \in \mathbb{R}^{2n}$, where $q_i(t)$ is the inventory and $S_i(t)$ the mid-price of asset i, and the control $u \in \mathbb{R}^n$ is the vector of liquidation rates. The dynamics are

$$\dot{q}_i = -u_i, \qquad \dot{S}_i = -\kappa_i u_i, \tag{6}$$

where $\kappa_i > 0$ is the permanent price-impact coefficient. Absorbing the cash process $\dot{X}_i = S_i u_i - \eta_i |u_i|^\gamma$ into the running cost via $X(T) - X(0) = \int_0^T \dot{X}\,\mathrm{d}t$, the objective becomes

$$\min_{u(\cdot)} \int_0^T L'(t, z, u)\,\mathrm{d}t + G'(z(T)), \tag{7}$$

with

$$L'(t, z, u) = \frac{1}{2}\sum_{i=1}^n \sigma_i^2 q_i^2 - \sum_{i=1}^n S_i u_i + \sum_{i=1}^n \eta_i (u_i^2 + \varepsilon)^{\gamma/2}, \tag{8}$$

$$G'(z(T)) = \alpha \sum_{i=1}^n q_i(T)^2, \tag{9}$$

where σ_i^2 is the variance of asset i, $\eta_i > 0$ the temporary impact coefficient, $\alpha > 0$ the terminal inventory penalty, and $\varepsilon > 0$ a smoothing constant. The three-state formulation tracking $X(t)$ explicitly is mathematically equivalent since $-X(0)$ is policy-independent, but including X in the OC state causes the inner fixed-point iteration to diverge; the analysis is in Section A.

4.2 Value network

We use the architecture of Gelphman et al. [2025],

$$\phi_\theta(t, z) = w^\top \mathrm{ResNet}([z; t]) + \tfrac{1}{2}[z; t]^\top A^\top A\,[z; t] + b^\top [z; t] + c, \tag{10}$$

a ResNet with an explicit quadratic-in-(z, t) head. We also tested a terminal-anchored wrapper $\phi_\theta(t, z) = G'(z) + (T - t)N_\theta(t, z)$, which enforces $\phi_\theta(T, z) = G'(z)$ exactly for any backbone weights because the prefactor $(T - t)$ vanishes at $t = T$. The results with the anchored form were nonphysical and are not included; all figures in this section use (10).

4.3 Experimental setup

Hyperparameters are listed in Section A.2. The network has three residual layers of width 50 with antiderivative-of-tanh activations. Training minimizes $\mathbb{E}_{x\sim\rho}[J_x(\theta)]$ via Adam. At each epoch a batch of initial inventories is sampled uniformly from $[0, q_{\max}]^n$ with S_0 fixed, rolled out under the current implicit policy, and the JFB surrogate gradient is computed as in Section 2. For the compute comparison an identical architecture is trained with full autodiff, unrolling the fixed-point iteration and trajectory in the computation graph.

4.4 Results

Multi-asset rollout. Figure 2 shows inventory and trading-rate trajectories for $n = 20$ assets. Inventories decrease monotonically and reach near-zero by $T = 2$, consistent with the terminal penalty. Trading rates are front-loaded and taper toward T, matching the qualitative Almgren-Chriss profile. All 20 assets are handled by a single shared value function with no modification to the pipeline.

Compute and memory. Figure 3 shows CPU memory against epoch for JFB (blue) and full autodiff (orange) on the 20-asset problem. Memory tells the same story as compute: JFB stays between 1500 and 2500 MB throughout, while full autodiff peaks near 7000 MB and settles around 5000 MB. The corresponding training loss vs. work units plot is deferred to the appendix (Figure 7). These numbers are consistent with the theoretical $O(m^2)$ versus $O(m^3)$ prediction of Gelphman et al. [2025].

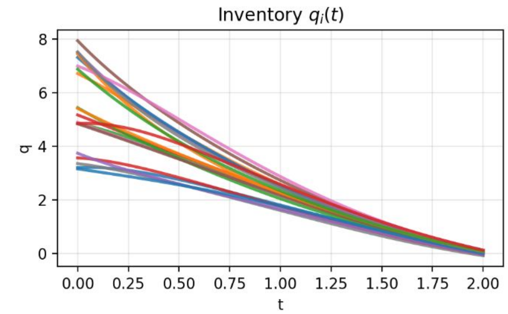
4.5 Discussion

The experiments confirm that the JFB pipeline learns a feedback policy for a 20-asset implicit optimal control problem without access to the closed-form solution during training, and that the compute and memory advantages reported in Gelphman et al. [2025] hold empirically on this benchmark.

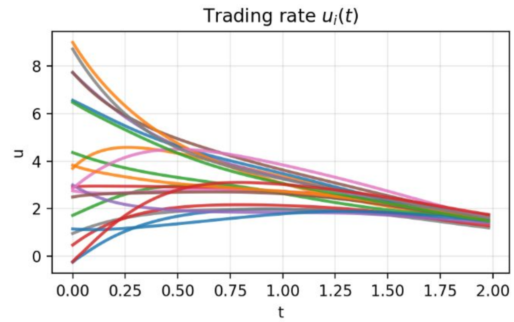
Our main addition is the contractivity analysis in Section A. The three-state formulation fails because the Hamiltonian curvature in u is $\nabla^2_{uu} H = -p_X A$, and the cash costate is pinned to $p_X \equiv -1$ by its terminal condition, making the curvature positive regardless of what the network learns. Dropping X from the OC state moves the curvature from a learned quantity to the problem constant 2η, which restores contractivity unconditionally. At $\gamma = 2$ this also gives the per-asset closed form $u_i^\star = (S_i + p_{q_i} + \kappa_i p_{S_i})/(2\eta_i)$ and exact one-step convergence at $\alpha_{\mathrm{fp}} = 1/(2\eta_i)$.

5 RL-like Extension

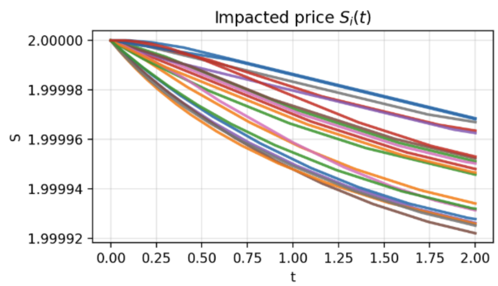
The pipeline of Gelphman et al. [2025] requires the dynamics f analytically in three places: the forward rollout, the Hamiltonian gradient $\nabla_u H = \nabla_u L + (\nabla_u f)^\top p$, and the adjoint equation $\dot{p} = -(\nabla_z f)^\top p - \nabla_z L$. We extend it to the setting where f is unknown: the agent observes transitions $(z_k, u_k) \mapsto z_{k+1}$ but has no access to f or its Jacobians. The costs L and G remain known. Every appearance of $\nabla_z f$ and $\nabla_u f$ is replaced by local estimates fitted from rollout data, leaving the value-function parameterization ϕ_θ, the fixed-point operator, and the JFB surrogate structurally intact.



(a) Inventory $q_i(t)$.



(b) Trading rates $u_i(t)$.



(c) Stock price $S(t)$.

(d) Cash account $X(t)$.

Figure 2: Multi-asset rollout for $n = 20$ assets under the JFB-trained policy.

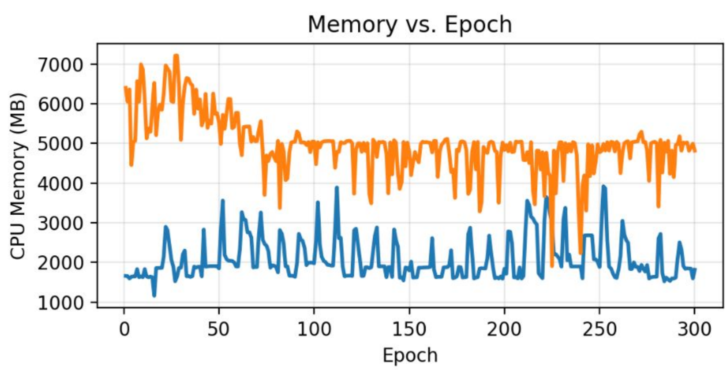


Figure 3: Peak CPU memory vs. epoch on the 20-asset problem.

5.1 Local Jacobian estimation

On a uniform grid $t_k = k\Delta t$, the unknown transition is

$$z_{k+1} = F(t_k, z_k, u_k) = z_k + \Delta t\, f + \mathcal{O}(\Delta t^2). \tag{11}$$

We fit an affine model $z_{k+1} \approx F_k[z_k; u_k; 1]$ at each step by batched recursive least squares with forgetting factor α_{rls}. Writing $F_k = [A_k^{\text{disc}} \mid B_k^{\text{disc}} \mid c_k]$, the continuous-time Jacobian estimates are

$$a_k = \frac{A_k^{\text{disc}} - I}{\Delta t} \approx \nabla_z f, \qquad b_k = \frac{(B_k^{\text{disc}})^\top}{\Delta t} \approx \nabla_u f. \tag{12}$$

We initialize $A_k^{\text{disc}} = I$ so that $a_k = 0$ before any data arrive, which prevents the adjoint from collapsing during warm-up. The full RLS update equations follow the augmented formulation of Eberhard et al. [2025].

5.2 Estimated operator and adjoint

Substituting b_k for $\nabla_u f$ in the fixed-point operator gives

$$\widehat{T}_{\theta,k}(u; z) = u + \alpha\big(\nabla_u L(t_k, z, u) + b_k \nabla_z \phi_\theta(t_k, z)\big), \tag{13}$$

the analogue of (5) with b_k in place of $\nabla_u f$. The fixed-point solve, Anderson acceleration, and control clamping are inherited unchanged from the deterministic pipeline.

The backward pass uses the estimated discrete adjoint

$$p_N = \nabla G(z_N), \qquad p_k = p_{k+1} + \Delta t\big(a_k^\top p_{k+1} + \nabla_z L(t_k, z_k, u_k)\big), \tag{14}$$

which depends on a_k but not on f. The derivation and its reconciliation with the discrete form $p_k = (A_k^{\text{disc}})^\top p_{k+1} + \Delta t\, \nabla_z L$ are in Section A.6.

5.3 JFB surrogate

After a no-gradient rollout that fixes the trajectory $(\bar{z}_k, \bar{u}_k)$ and detached costates $\widehat{p}_{k+1}$, we backpropagate the scalar surrogate

$$S(\theta) = \sum_{k=0}^{N-1} \big\langle \widehat{T}_{\theta,k}(\bar{u}_k; \bar{z}_k),\ \Delta t\big(\nabla_u L(t_k, \bar{z}_k, \bar{u}_k) + b_k^\top \widehat{p}_{k+1}\big)\big\rangle, \tag{15}$$

where only $\widehat{T}_{\theta,k}$ depends on θ through ϕ_θ. Its gradient reproduces the estimated JFB direction by construction, since $\bar{u}_k$, b_k, and $\widehat{p}_{k+1}$ carry no gradient.

5.4 Exploration

A converged policy produces nearly deterministic controls, which starves the RLS estimator of signal for identifying b_k. We therefore optionally run an exploration rollout with additive control noise $u_k \leftarrow u_k + \sigma\varepsilon_k$, $\varepsilon_k \sim \mathcal{N}(0, I)$, used only to update the estimates (12), and a separate clean rollout that defines the trajectory, costs, adjoint, and surrogate. The scale σ is decayed across epochs; at $\sigma = 0$ the two rollouts collapse to one.

5.5 Conjecture

The estimated gradient makes two approximations beyond those in Gelphman et al. [2025]: substituting a_k for $\nabla_z f$ in the adjoint and b_k for $\nabla_u f$ in the operator and surrogate. Whether the resulting direction remains a descent direction is an open question. Fusing the contractivity and conditioning hypotheses of Gelphman et al. [2025] with the bounded-estimation-error hypothesis of Eberhard et al. [2025] suggests it should, but we have not established a proof.

Conjecture 1 (Descent under estimation) *Under the contractivity, conditioning, and time-regularity assumptions of Gelphman et al. [2025] and the Jacobian-estimation bounds of Assumption 2 of Eberhard et al. [2025], there exist $\mu, \nu > 0$ such that the estimated gradient satisfies*

$$\Big\langle -\widehat{\tfrac{\mathrm{d}J_x}{\mathrm{d}\theta}}, -\tfrac{\mathrm{d}J_x}{\mathrm{d}\theta}\Big\rangle \ge \mu \big\|\tfrac{\mathrm{d}J_x}{\mathrm{d}\theta}\big\|^2, \qquad \big\|\widehat{\tfrac{\mathrm{d}J_x}{\mathrm{d}\theta}}\big\| \le \nu \big\|\tfrac{\mathrm{d}J_x}{\mathrm{d}\theta}\big\|.$$

5.6 Algorithm

5.7 Merton portfolio

As a first instantiation we apply the pipeline to a Merton-type portfolio problem. Wealth evolves by $\dot{W} = rW + \pi(\mu - r)W$, where π is the risky-asset fraction and the drift μ and risk-free rate r are hidden from the agent. The known costs are

$$L(t, W, \pi) = \lambda\big(e^{\pi^2} - 1\big), \qquad G(W) = -\log(W/W_{\text{ref}}). \tag{16}$$

The optimality condition $-p(\mu - r)W - 2\lambda\pi e^{\pi^2} = 0$ is transcendental in π, so the Hamiltonian is implicit. With $n = m = 1$ the Jacobian estimates are scalars, $a_k \approx r + \bar{\pi}_k(\mu - r)$ and $b_k \approx (\mu - r)\bar{W}_k$,

Algorithm 1 JFB training with estimated Jacobians (one epoch)

Require: ϕ_θ, step α, rate η, RLS params $(\alpha_{\text{rls}}, q_0)$, exploration scale σ, horizon N.

```
sample z_0 ~ ρ
if σ > 0 then                                                    ▷ exploration rollout, no grad
    for k = 0, ..., N−1: u_k ← FP(T̂_{θ,k}; z_k) + σε_k; step env; update a_k, b_k via (12)
end if
clean rollout (no grad): u_k ← FP(T̂_{θ,k}; z_k), step env, accumulate L; if σ = 0 update a_k, b_k here
adjoint (no grad): p_N ← ∇G(z_N); then (14) for k = N−1, ..., 0
JFB step: backprop S(θ); θ ← θ − η∇_θ S; decay σ
```

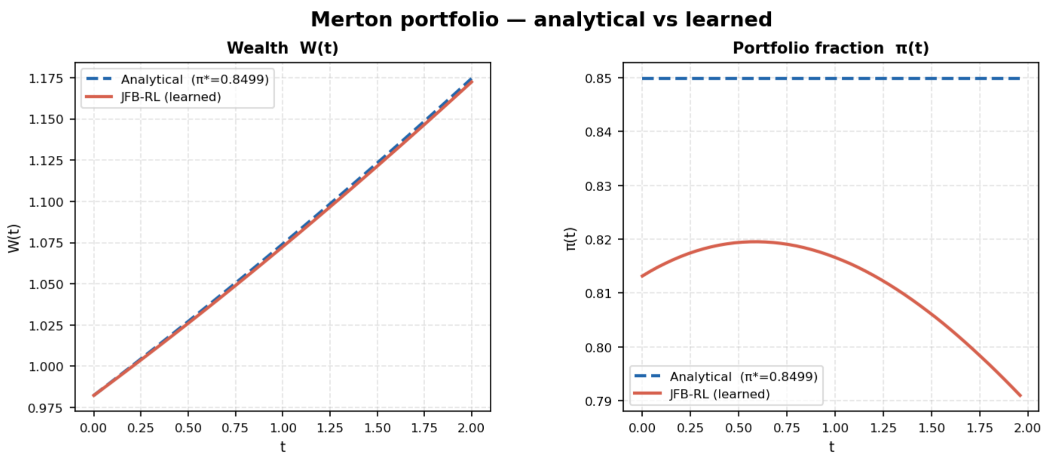


Figure 4: Merton portfolio: learned policy (red) against the analytical optimum (dashed blue). Left: wealth $W(t)$ closely tracks the reference. Right: allocation $\pi(t)$ stays near but does not reproduce the constant optimum $\pi^\star = 0.85$.

making this the most favorable possible setting for RLS identifiability. We treat it as a sanity check rather than a stringent test: once (μ, r) are revealed, direct shooting gives an exact ground-truth feedback law.

Training converges smoothly over 200 epochs. Figure 4 shows the learned policy against the analytical optimum. Wealth $W(t)$ closely tracks the reference curve. The learned allocation $\pi(t)$ stays near 0.79–0.82 rather than the constant optimum $\pi^\star = 0.85$ and drifts slightly toward the terminal time, but the cost mismatch is small, consistent with a flat objective near the optimum.

5.8 Van der Pol stabilization

Our main benchmark is stabilization of a Van der Pol oscillator with state $z = (x_1, x_2)$, scalar control $u \in [-3, 3]$, and dynamics

$$\dot{x}_1 = x_2, \qquad \dot{x}_2 = (1 - x_1^2)x_2 - x_1 + g(x_1)u, \tag{17}$$

over horizon $T = 3$, with terminal cost $G = x_1^2 + x_2^2$ and running cost $L = x_1^2 + x_2^2$ plus a control penalty. The agent does not know (17) and estimates Jacobians by RLS. We test three variants of increasing difficulty. The *standard* variant ($g \equiv 1$, quadratic penalty $\frac{1}{2}u^2$) has an explicit maximizer $u^\star = -p_2$ and serves only as a sanity check. The *hard* variant replaces the penalty by $\lambda_u(e^{u^2} - 1)$, making $\nabla_u H = 0$ transcendental and genuinely requiring the implicit solver. The *hard-gain* variant additionally sets $g(x_1) = 1 + \beta \tanh(x_1)$, so $\nabla_u f$ varies along the trajectory and directly stresses the RLS estimator.

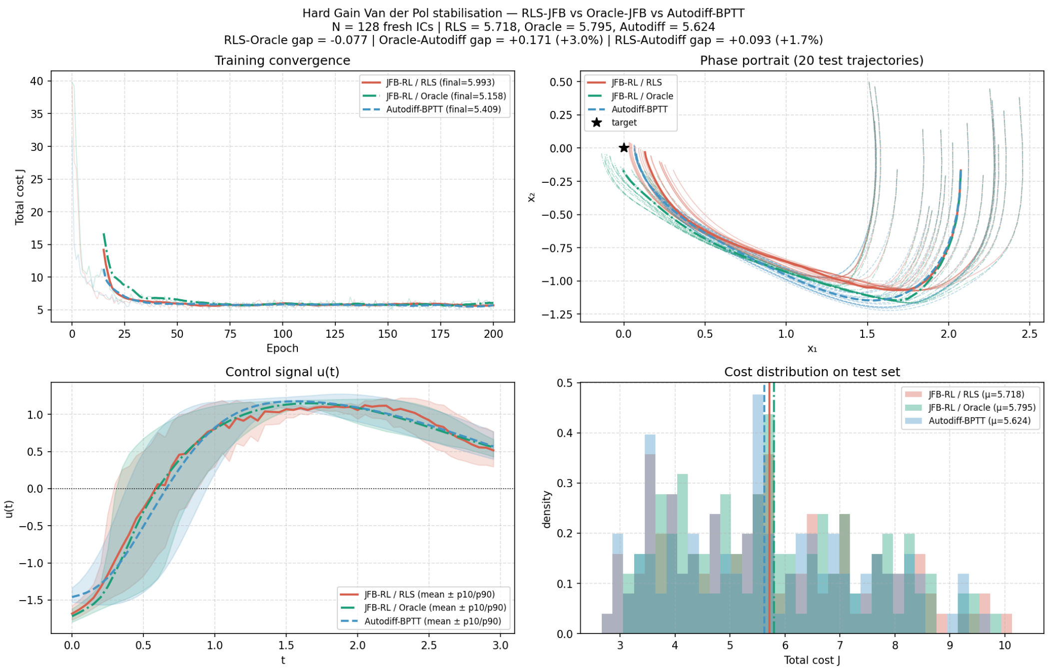


Figure 5: Hard-gain Van der Pol on $N = 128$ held-out initial conditions. JFB-RL with RLS estimates (red), oracle Jacobians (green), and BPTT (blue). Top left: training convergence. Top right: phase portraits of 20 test trajectories. Bottom left: mean control with $p10/p90$ band. Bottom right: test-set cost distribution. Mean costs: 5.718 (RLS), 5.795 (oracle), 5.624 (BPTT).

For each variant we compare three gradients on an identical network: JFB-RL with RLS-estimated Jacobians, the same gradient with oracle Jacobians, and BPTT. The oracle variant isolates the JFB approximation error; any gap between RLS and oracle is attributable to estimation alone.

Figure 5 reports the hard-gain variant on $N = 128$ held-out initial conditions. Mean test costs are 5.718 (RLS), 5.795 (oracle), and 5.624 (BPTT). The oracle-versus-BPTT gap of $+3.0\%$ measures the JFB approximation cost. The RLS-versus-oracle gap of -0.077 is within run-to-run variation, indicating no measurable degradation from Jacobian estimation. The pipeline trails exact BPTT by 1.7% while using neither the dynamics nor their Jacobians.

The RLS-versus-oracle gap also bears on Conjecture 1: a failure of the estimation hypothesis would show as a positive gap widening with horizon and non-monotone training. We see neither, which is consistent with the conjecture on this instance but does not establish it.

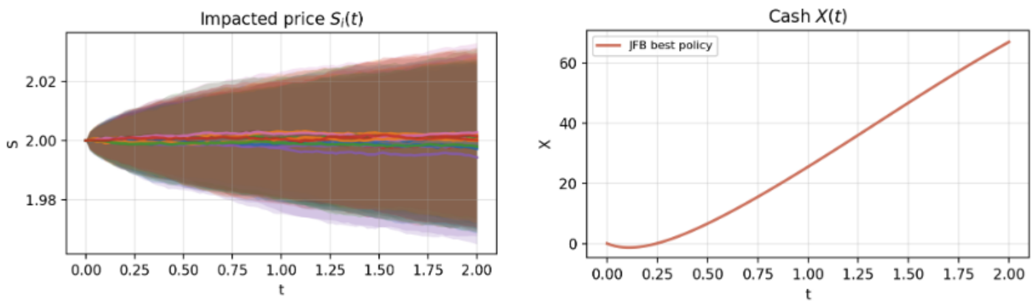
6 Stochastic price dynamics

In realistic execution settings the mid-price is not deterministic: it fluctuates around the impact-driven drift due to exogenous market noise. We extend the liquidation model by adding independent Brownian motions to the price dynamics,

$$\mathrm{d}S_i = -\kappa_i u_i \,\mathrm{d}t + \sigma_S \,\mathrm{d}W_i, \tag{18}$$

while keeping the inventory dynamics $\dot{q}_i = -u_i$ and the running and terminal costs unchanged. The state is now a stochastic process $z(t) = (q(t), S(t))$ and the objective becomes an expectation over price paths,

$$\min_{u(\cdot)} \mathbb{E}\left[\int_0^T L'(t, z, u)\,\mathrm{d}t + G'(z(T))\right]. \tag{19}$$



(a) Impacted price $S_i(t)$ across sample paths. (b) Accumulated cash $X(t)$ vs. deterministic baseline.

Figure 6: Stochastic liquidation rollout under (18). The fan in (a) reflects the Brownian perturbation across sample paths. Cash in (b) exceeds the deterministic baseline due to adaptive timing by the closed-loop policy.

Effect on the value function and training. The value function now satisfies the stochastic HJB equation, which acquires an Itô correction term relative to the deterministic case,

$$-\partial_t \phi + H(t, z, \nabla_z \phi) - \tfrac{1}{2}\sigma_S^2 \sum_i \partial^2_{S_i S_i} \phi = 0. \tag{20}$$

Because the diffusion in (18) is additive and control-independent, the Hamiltonian H and the optimality condition $\nabla_u H = 0$ are unchanged: noise enters only through the drift of the costate, not through the control coupling. By certainty equivalence, the stochastic-optimal feedback law coincides with the deterministic one, and we do not modify the training pipeline or enforce the Itô residual explicitly. In practice we train by minimizing the sample-average cost over Euler-Maruyama rollouts,

$$S_{i,k+1} = S_{i,k} - \kappa_i u_{i,k} \Delta t + \sigma_S \sqrt{\Delta t}\, \varepsilon_{i,k}, \quad \varepsilon_{i,k} \sim \mathcal{N}(0, 1), \tag{21}$$

with the fixed-point operator and JFB surrogate applied at each step exactly as in the deterministic case. The contractivity analysis of Section A carries over identically: the cash dynamics $\mathrm{d}X_i = S_i u_i \,\mathrm{d}t - \eta_i |u_i|^\gamma \mathrm{d}t$ carry no diffusion term, so $p_X \equiv -1$ still holds and the reduced formulation remains the correct one.

Results. Figure 6 shows representative trajectories. Terminal inventories reach near-zero across all sample paths, confirming the policy remains feasible under price noise. Accumulated cash $X(T)$ exceeds the deterministic baseline. The mechanism is that the JFB policy is a closed-loop feedback law $u^\star(t, q(t), S(t))$ conditioning on the realized price at each step. When S_i rises due to a favorable noise draw the policy sells more aggressively, capturing the elevated price; when S_i falls it slows down. This generates a positive covariance $\mathrm{Cov}(S_i(t), u_i(t)) > 0$, and since

$$\mathbb{E}[X(T)] = \int_0^T \sum_i \big(\mathbb{E}[S_i]\, \mathbb{E}[u_i] + \mathrm{Cov}(S_i, u_i)\big)\, \mathrm{d}t - \text{impact costs}, \tag{22}$$

the covariance term is a timing premium unavailable to any open-loop or deterministic policy, for which $\mathrm{Cov} = 0$ by construction. The gain is not purchased at the cost of residual inventory: $q_i(T) \approx 0$ across all paths, so the terminal penalty remains negligible.

Scope and limitations. This experiment probes robustness of a deterministically trained policy under stochastic deployment, not optimality under stochastic training. Certainty equivalence holds here because the diffusion is additive and control-independent, but would break under multiplicative or control-dependent noise, e.g. $\mathrm{d}S_i = -\kappa_i u_i \,\mathrm{d}t + \sigma_i(u_i)\,\mathrm{d}W_i$, where the Itô correction in (20) depends on u and can no longer be ignored. Training on such dynamics would require either enforcing the stochastic HJB residual via a Hutchinson trace estimator for $\partial^2_{SS}\phi_\theta$, or an FBSDE-based approach [Li et al., 2024]; we leave this to future work.

References

Robert Almgren and Neil Chriss. Optimal execution of portfolio transactions. *Journal of Risk*, 3(2): 5–39, 2001. doi: 10.21314/JOR.2001.041.

Brandon Amos, Ivan Jimenez, Jacob Sacks, Byron Boots, and J. Zico Kolter. Differentiable MPC for end-to-end planning and control. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

Onno Eberhard, Claire Vernade, and Michael Muehlebach. A Pontryagin perspective on reinforcement learning. In *Proceedings of the Seventh Annual Learning for Dynamics & Control Conference*, volume 283 of *Proceedings of Machine Learning Research*, pages 233–244. PMLR, 2025. URL `https://arxiv.org/abs/2405.18100`.

Wendell H. Fleming and Halil Mete Soner. *Controlled Markov Processes and Viscosity Solutions*. Springer, 2nd edition, 2006.

Eric Gelphman, Deepanshu Verma, Nicole Tianjiao Yang, Stanley Osher, and Samy Wu Fung. End-to-end training of high-dimensional optimal control with implicit hamiltonians via jacobian-free backpropagation. *arXiv preprint arXiv:2510.00359*, 2025. doi: 10.48550/arXiv.2510.00359. URL `https://arxiv.org/abs/2510.00359`.

Eric Gelphman, Deepanshu Verma, Nicole Tianjiao Yang, Stanley Osher, and Samy Wu Fung. On the convergence of Jacobian-free backpropagation for optimal control problems with implicit Hamiltonians. *arXiv preprint arXiv:2602.00921*, 2026. URL `https://arxiv.org/abs/2602.00921`.

Michael Hertneck, Johannes Hölzell, Sebastian Trimpe, and Frank Allgöwer. Learning an approximate model predictive controller with guarantees. *IEEE Control Systems Letters*, 2(3):543–548, 2018. doi: 10.1109/LCSYS.2018.2843682.

Xingjian Li, Deepanshu Verma, and Lars Ruthotto. A neural network approach for stochastic optimal control. *SIAM Journal on Scientific Computing*, 46(5):C535–C556, 2024. doi: 10.1137/23M155832X.

Derek Onken, Levon Nurbekyan, Xingjian Li, Samy Wu Fung, Stanley Osher, and Lars Ruthotto. A neural network approach for high-dimensional optimal control applied to multiagent path finding. *IEEE Transactions on Control Systems Technology*, 31(1):235–251, 2023. doi: 10.1109/TCST.2022.3179670.

Lev S. Pontryagin, Vladimir G. Boltyanskii, Revaz V. Gamkrelidze, and Evgenii F. Mishchenko. *The Mathematical Theory of Optimal Processes*. Interscience Publishers, 1962.

Lars Ruthotto, Stanley J. Osher, Wuchen Li, Levon Nurbekyan, and Samy Wu Fung. A machine learning framework for solving high-dimensional mean field game and mean field control problems. *Proceedings of the National Academy of Sciences*, 117(17):9183–9193, 2020. doi: 10.1073/pnas.1922204117.

Samy Wu Fung, Howard Heaton, Qiuwei Li, Daniel McKenzie, Stanley Osher, and Wotao Yin. JFB: Jacobian-free backpropagation for implicit networks. *Proceedings of the AAAI Conference on Artificial Intelligence*, 36(4):3849–3857, 2022. doi: 10.1609/aaai.v36i4.20296.

Ming Xu, Timothy L. Molloy, and Stephen Gould. Revisiting implicit differentiation for learning problems in optimal control. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

A Contractivity and the reduced formulation

A.1 The contractivity obstruction in the full formulation

The JFB pipeline requires the fixed-point operator

$$T_\theta(u;\,t,z) = u + \alpha_{\text{fp}}\,\nabla_u H(t,z,\nabla_z\phi_\theta(t,z),u) \tag{23}$$

to be a Γ_c-contraction in u, with $\Gamma_c < 1$. The contraction constant is

$$\Gamma_c = \big\|I + \alpha_{\text{fp}}\,\nabla^2_{uu}H\big\|, \tag{24}$$

so we need $\nabla^2_{uu}H \prec 0$, i.e. H strictly concave in u.

In the full three-state formulation the state is $z = (q,S,X)$ and the Hamiltonian is

$$H = -p_q(-u) - p_S(-\kappa u) - p_X(Su - \eta|u|^\gamma) - \tfrac{1}{2}\sigma^2 q^2, \tag{25}$$

where (p_q,p_S,p_X) are the costates. The curvature of H in u is

$$\nabla^2_{uu}H = -p_X\,\eta\gamma(\gamma-1)|u|^{\gamma-2} =: -p_X\,A, \qquad A > 0. \tag{26}$$

The adjoint terminal condition for cash is $p_X(T) = -\partial G/\partial X = -1$ (the marginal value of a dollar of terminal cash is exactly one), and p_X satisfies $\dot{p}_X = -\partial H/\partial X = 0$, so

$$p_X(t) \equiv -1 \quad \forall\, t \in [0,T]. \tag{27}$$

Substituting (27) into (26) gives

$$\nabla^2_{uu}H = +A > 0, \tag{28}$$

so H is *convex* in u and $\Gamma_c = \|I + \alpha_{\text{fp}}A\| > 1$. The contraction condition is violated: the inner fixed-point iteration diverges regardless of the step size $\alpha_{\text{fp}} > 0$.

The root cause is that the stabilizing curvature $-p_X A$ depends on the *learned* costate $p_X = \partial_X\phi_\theta$. At random initialization p_X is arbitrary, and even at optimum the structural identity $p_X \equiv -1$ is only recovered asymptotically. The full formulation therefore places the contraction burden on a quantity the network must learn, making training circular and unstable in practice.

A.1.1 The reduced formulation restores contractivity

Eliminating X from the OC state removes p_X from $\nabla^2_{uu}H$ entirely, replacing it with a problem constant. In the reduced formulation with $z = (q,S)$ and L' as in (8), the Hamiltonian is

$$H' = p_q(-u) + p_S(-\kappa u) - \tfrac{1}{2}\sigma^2 q^2 + Su - \eta(u^2+\varepsilon)^{\gamma/2}, \tag{29}$$

and its curvature in u is

$$\nabla^2_{uu}H' = -\eta\gamma(\gamma-1)(u^2+\varepsilon)^{\gamma/2-1} \prec 0 \quad \text{for all } \gamma > 1,\ \eta > 0. \tag{30}$$

This is a *problem constant*, independent of any learned quantity, so $\Gamma_c = \|I + \alpha_{\text{fp}}\nabla^2_{uu}H'\| < 1$ is guaranteed for $0 < \alpha_{\text{fp}} < 2/(\eta\gamma(\gamma-1)\max|u|^{\gamma-2})$.

Closed form at $\gamma = 2$. For $\gamma = 2$ the optimality condition $\nabla_u H' = 0$ is linear in u, giving per asset

$$u_i^* = \frac{S_i + p_{q_i} + \kappa_i\,p_{S_i}}{2\eta_i}. \tag{31}$$

With the choice $\alpha_{\text{fp}} = 1/(2\eta_i)$, one application of T_θ starting from any $u^{(0)}$ yields u^* exactly: $T_\theta(u^{(0)}) = u^*$. The fixed-point solver converges in a *single iteration*, which we verify empirically.

Terminal-anchored value network. To enforce the terminal conditions $p_q(T) = 2\alpha q(T)$ and $p_S(T) = 0$ exactly, we parameterize the value function as

$$\phi_\theta(t,z) = G'(z) + (T-t)\,N_\theta(t,z), \tag{32}$$

where N_θ is a neural network backbone. Because the prefactor $(T-t)$ vanishes at $t = T$, $\phi_\theta(T,z) = G'(z)$ holds identically for any backbone weights, and hence $\nabla_z\phi_\theta(T,z) = \nabla_z G'(z)$ as well. The boundary conditions are therefore satisfied at initialization and preserved throughout training without any constraint on N_θ itself.

Table 1: Hyperparameters for the Almgren–Chriss reproduction.

Parameter	Symbol	Single asset	20 assets
Horizon	T	2.0	2.0
Time steps	N_t	40	40
Assets	n	1	20
Volatility	σ_i	0.1	0.1
Perm. impact	κ_i	0.1	0.1
Temp. impact	η_i	1.0	1.0
Impact exponent	γ	2	2
Smoothing	ε	10^{-4}	10^{-4}
Terminal penalty	α	5.0	5.0
FP step size	α_{fp}	0.5	0.5
FP iterations (detached)	K	20	20
FP iterations (on-graph)	K'	3	3
Initial inventory	q_0	$[0, 8]$	$[0, 8]^n$
Initial price	S_0	10.0	10.0
Batch size	B	64	64
Learning rate	η_θ	10^{-3}	10^{-3}
Epochs	—	500	500

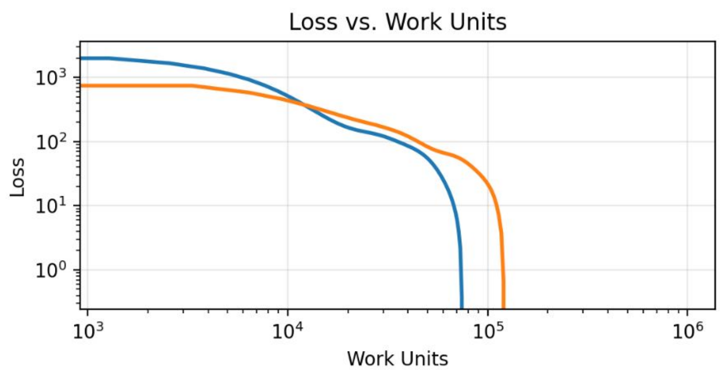


Figure 7: Training loss vs. work units on the 20-asset problem (JFB vs. full autodiff).

Cash as a passive observer. The cash process $X(t)$ is integrated forward alongside the OC dynamics (6) for diagnostic purposes, but it never enters the loss, the costate recursion, or the fixed-point operator.

A.2 Hyperparameters for the Almgren–Chriss reproduction

A.3 Additional compute plots

A.4 Background details

This appendix records the standard equations that are summarized in Section 2.

A.4.1 Contraction mapping

Let $T : U \to U$ be a map on a normed space $(U, \|\cdot\|)$. We say that T is a contraction if there exists a constant $\Gamma \in [0, 1)$ such that

$$\|T(u) - T(v)\| \le \Gamma \, \|u - v\| \qquad \text{for all } u, v \in U. \tag{33}$$

Equivalently, T is Lipschitz with constant strictly smaller than one. In this case the fixed-point iteration $u^{(k+1)} = T(u^{(k)})$ converges to the unique fixed point.

A.4.2 Pontryagin maximum principle

For the problem (1), introduce a costate $p(t)$ and the Hamiltonian

$$H(t, z, p, u) = -p^\top f(t, z, u) - L(t, z, u). \tag{34}$$

The PMP conditions give

$$\dot{z}(t) = -\nabla_p H(t, z, p, u^\star) = f(t, z, u^\star), \tag{35}$$

$$\dot{p}(t) = \nabla_z H(t, z, p, u^\star) = -(\nabla_z f)^\top p - \nabla_z L, \tag{36}$$

$$u^\star(t) \in \arg\max_u H(t, z(t), p(t), u), \tag{37}$$

with $z(0) = x$ and $p(T) = \nabla G(z(T))$.

A.4.3 HJB and the PMP–value link

The value function $\phi(t, z)$ satisfies

$$-\partial_t \phi(t, z) + H(t, z, \nabla_z \phi(t, z)) = 0, \quad \phi(T, z) = G(z). \tag{38}$$

Along an optimal trajectory $z^\star(t)$ one has

$$p(t) = \nabla_z \phi(t, z^\star(t)). \tag{39}$$

A.4.4 Implicit Hamiltonians

When $u \mapsto H(t, z, p, u)$ has no closed-form maximizer, the first-order optimality condition

$$\nabla_u H(t, z, p, u) = -(\nabla_u f)^\top p - \nabla_u L(t, z, u) = 0 \tag{40}$$

must be solved numerically.

A.5 Derivation of the PMP conditions

We sketch the standard calculus-of-variations argument. Define the augmented functional

$$\mathcal{J}(z, u, p) = \int_0^T \Big(L(t, z, u) + p(t)^\top (\dot{z} - f(t, z, u)) \Big) \, \mathrm{d}t + G(z(T)). \tag{41}$$

Taking first variations in z and integrating $p^\top \dot{z}$ by parts gives the adjoint equation (36) with terminal condition $p(T) = \nabla G(z(T))$. Variations in u yield stationarity of $u \mapsto H(t, z, p, u)$, which leads to (37).

A.6 Derivation of the discrete adjoint recursion

We derive the backward recursion (14) used in the estimated JFB gradient, and reconcile the two forms in which it appears: the discrete form $p_k = (A_k^{\mathrm{disc}})^\top p_{k+1} + \Delta t \, \nabla_z L$ and the continuous-time form $p_k = p_{k+1} + \Delta t \, (a_k^\top p_{k+1} + \nabla_z L)$ that the implementation computes. The derivation is the standard discrete adjoint / KKT argument; we record it because the substitution of the estimated Jacobians and the convention relating a_k to A_k^{disc} are the only places the RL setting differs from the deterministic one, and both are easy to misread.

Discretized problem and Lagrangian. Fix the uniform grid $t_k = k\Delta t$, $k = 0, \dots, N$, and let $z_{k+1} = F(t_k, z_k, u_k)$ denote the one-step transition map of (11). The discretized optimal-control problem is

$$\min_{u_{0:N-1},\, z_{0:N}} \sum_{k=0}^{N-1} L(t_k, z_k, u_k) \, \Delta t + G(z_N) \qquad \text{s.t.} \qquad z_{k+1} = F(t_k, z_k, u_k), \;\; z_0 = x. \tag{42}$$

Introducing multipliers $p_{1:N}$ for the N dynamics constraints, the Lagrangian is

$$\mathcal{L} = \sum_{k=0}^{N-1} L(t_k, z_k, u_k) \, \Delta t + G(z_N) + \sum_{k=0}^{N-1} p_{k+1}^\top \big(F(t_k, z_k, u_k) - z_{k+1} \big). \tag{43}$$

Stationarity in the states. The state z_k for an interior index $1 \le k \le N-1$ appears in (43) in exactly two terms: the running cost $L(t_k, z_k, u_k)\Delta t$ and, through $F(t_k, z_k, u_k)$, the constraint term $p_{k+1}^\top F(t_k, z_k, u_k)$; it also appears as the target $-p_k^\top z_k$ coming from the $(k-1)$-th constraint. Collecting these, $\partial \mathcal{L}/\partial z_k = 0$ gives

$$0 = \Delta t\, \nabla_z L(t_k, z_k, u_k) + (\nabla_z F_k)^\top p_{k+1} - p_k, \tag{44}$$

where $\nabla_z F_k := \partial F(t_k, z_k, u_k)/\partial z_k$. Rearranging,

$$\boxed{p_k = (\nabla_z F_k)^\top p_{k+1} + \Delta t\, \nabla_z L(t_k, z_k, u_k)} \qquad k = N-1, \dots, 1. \tag{45}$$

Terminal condition. The terminal state z_N appears only in $G(z_N)$ and in the last constraint term $-p_N^\top z_N$. Setting $\partial \mathcal{L}/\partial z_N = 0$ yields

$$p_N = \nabla G(z_N). \tag{46}$$

Optimality in the controls. For completeness, $\partial \mathcal{L}/\partial u_k = 0$ gives

$$0 = \Delta t\, \nabla_u L(t_k, z_k, u_k) + (\nabla_u F_k)^\top p_{k+1} \iff \nabla_u L + (\nabla_u f_k)^\top p_{k+1} = 0, \tag{47}$$

which is the discrete Pontryagin first-order condition. Under the identification $p_{k+1} \approx \nabla_z \phi_\theta(t_{k+1}, z_{k+1})$ it is exactly the condition $\nabla_u H = 0$ enforced by the fixed-point operator, linking the adjoint multipliers to the costate used inside $\widehat{T}_{\theta,k}$.

From the true map to estimated Jacobians. Equations (45)–(46) are exact for the true transition map F. In the RL setting $\nabla_z F_k$ is unavailable, and we replace it by the RLS estimate A_k^{disc} of the discrete one-step state Jacobian:

$$p_k = (A_k^{\text{disc}})^\top p_{k+1} + \Delta t\, \nabla_z L(t_k, z_k, u_k), \qquad p_N = \nabla G(z_N). \tag{48}$$

Reconciliation with the continuous-time form. The estimator stores the continuous-time Jacobian $a_k = (A_k^{\text{disc}} - I)/\Delta t$ (Equation (12)), equivalently $A_k^{\text{disc}} = I + \Delta t\, a_k$. Substituting into (48),

$$p_k = (I + \Delta t\, a_k)^\top p_{k+1} + \Delta t\, \nabla_z L = p_{k+1} + \Delta t\big(a_k^\top p_{k+1} + \nabla_z L(t_k, z_k, u_k)\big), \tag{49}$$

which is the form (14) computed in the implementation. The two are algebraically identical given $A_k^{\text{disc}} = I + \Delta t\, a_k$; they differ only in whether the stored quantity is the discrete map Jacobian or its continuous-time counterpart. We flag this because a naive reading that treats the stored a_k as if it were A_k^{disc} — i.e. computing $p_k = a_k^\top p_{k+1} + \Delta t\, \nabla_z L$ — drops the p_{k+1} term and produces an incorrect adjoint; the identity-initialization of A_k^{disc} noted in Section 5.1is the corresponding safeguard at the start of training, ensuring $a_k = 0$ (rather than $-I/\Delta t$) gives $p_k = p_{k+1} + \Delta t\, \nabla_z L$ before any data are seen.

Continuous-time limit. Taking $\Delta t \to 0$ in (49) recovers the costate ODE of the Pontryagin system, $\dot{p} = -(\nabla_z f)^\top p - \nabla_z L$ with $p(T) = \nabla G(z(T))$, confirming that (14) is a consistent discretization of the adjoint equation when $a_k \to \nabla_z f$.